Trigonometric integrals (7.2)

This section covers methods to evaluate integrals of the forms

$$\int \sin^m x \cos^n x \, dx \qquad \int \tan^m x \sec^n x \, dx$$

where m, n are nonnegative integers.

The main tools we will use are trigonometric identities and the substitution rule.

Useful trig identities:

$\sin^2 x + \cos^2 x = 1$

$\tan^2 x + 1 = \sec^2 x$

$\sin 2x = 2\sin x \cos x$

$\cos 2x = 1 - 2\sin^2 x$

$\cos 2x = 2\cos^2 x - 1$

$\sin x \cos x = \frac{1}{2}\sin 2x$

$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$

$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$

$\int \sin^m x \cos^n x$ where **m is odd** (n anything)

Use $\sin^2 x = 1 - \cos^2 x$ to rewrite $\sin^m x$ in terms of $\cos x$, keeping one factor of $\sin x$.

Substitute $u = \cos x$.

$$\int \sin^3 x \cos^2 x \, dx$$

$$\int_0^{\pi/6} \sin^5 x \, dx$$

$\int \sin^m x \cos^n x$ where n is odd (m anything)

Use $\cos^2 x = 1 - \sin^2 x$ to rewrite $\cos^n x$ in terms of $\sin x$, keeping one factor of $\cos x$.

Substitute $u = \sin x$.

(If both m and n are odd, you can use either method.)

$$\int_0^{\pi/6} \cos^3 x \, dx$$

$$\int \sin^2(\pi x)\cos^3(\pi x)\,dx$$

$\int \sin^m x \cos^n x$ where m, n are both even

Use the identities

$$\sin x \cos x = \frac{1}{2} \sin 2x$$

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

to reduce powers.

$$\int \cos^2 x \, dx$$

$$\int \sin^2 x \cos^2 x \, dx$$

$$\int \tan^m x \sec^n x \, dx$$

$\frac{d}{dx} \tan x = \sec^2 x$ $\qquad$ $\frac{d}{dx} \sec x = \tan x \sec x$

$\tan^2 x + 1 = \sec^2 x$

Try to do one of the following

(a) Convert everything to $\tan x$ except one factor of $\sec^2 x$, substitute $u = \tan x$.

(b) Convert everything to $\sec x$ except one factor of $\tan x \sec x$, substitute $u = \sec x$.

$$\int_0^{\pi/4} \tan^3 x \sec^4 x \, dx$$